

Recruitment Scam Detection Using Gated Recurrent Unit

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Abstract—Recruitment advertisements including fake or fraudulent job advertisements is delivered to any job seekers and lead them to lose their money or personal information by frauds. With mass digitization and internet access scammers are coming up with new ideas and recruitment fraud is one of them. By sending false recruitment emails, SMS texts, fake websites, fake social media profiles, fake job posts, online recruiting services such as LinkedIn, or unsolicited emails purporting to be from well-known organizations are common methods of recruitment fraud. To identify fake or fraudulent job advertisements, we have proposed a one layer gated recurrent unit (GRU) model that can classify scams and real recruitments. We also evaluate the efficiency and achieved 93.51% of AUC score on Employment Scam Aegean Dataset (EMSCAD) dataset.

Index Terms—Machine Learning, NLP, GRU, Word Embedding, Recruitment scam.

I. INTRODUCTION

In the present day, the development in modern technology has opened a massive probability of new and numerous jobs for job seekers. With the help of the internet, different organizations give online advertisements to hire people for their vacant posts. From those job offers, job seekers are able to find out their choices relying on their time, qualification, experience, suitability, and so on which is very helpful for one to find a suitable job within a short time. In spite of those advantages, it also has some drawbacks one of them is a recruitment scam that causes harassment to job seekers.

An online recruitment scam is a type of spam message that is sent via hackers targeting to get users' sensitive information. Such fraud

advertisements come not only from domestic but also international senders. Moreover, some innocent people without knowing anything, follow the instructions and give them their sensitive information. Afterward, the hacker uses those information to blackmail those people. Sometimes some fraud online recruitment even told job seekers that they were hired and then, they got resigned from their previous job. But ultimately, were ended up being scammed.

To mitigate this kind of harassment of job seekers online, here in this paper we proposed a 1-layered gated recurrent unit (GRU) model. This model can detect which job recruitment is real and which job recruitment is a scam.

II. LITERATURE REVIEW

Several types of research works are done to detect recruitment scams. Vidros [1] et al. in their work analyzed the possible aspects of employment scams and using several classification algorithms experimenting with EMSCAD dataset the results show that conjunction with metadata in text mining is able to provide a basic substructure for the related algorithms which are used job scam detection.

Alghamdi [2] et al. designed a model which can detect fake or fraud disclosure in online job advertisement environments. They used two data mining algorithms, for feature selections they use a support vector machine (SVM) and Random Forest is used as an ensemble classifier.

Huynh [3] et al. in their work the TextCNN model, Bi-GRU-LSTM-CNN and BiGRU-CNN

model, and many other complex models were implemented with various word embeddings which are pre-trained with text datasets to solve the IT job prediction.

Dutta [4] et al. proposed an automated tool using machine learning based on classification techniques to detect job scams. Several machine learning algorithms are proposed as countermeasures.

Habiba [5] et al. experimented on EMSCAD dataset using different classification algorithms and data mining techniques like KNN, naive bayes classifier, support vector machine, decision tree, random forest classifier, multilayer perceptron, and deep neural network to predict recruitment scam.

A Kumar [6] designed two self-attention enhanced GRU networks in his work. One of them is with vanilla RNN architecture and the other one is with Bidirectional architecture, for classifying the fake job from real ones. Experimenting on Employment Scam Aegean Dataset the bidirectional self-attention GRU model shows the best result with an accuracy of 97.40 compared to other machine learning algorithms.

After reviewing all those research works, it shows that methods based on GRUs and LSTM using deep learning are more popular and effective methods for classifying online job ads scams.

III. METHODOLOGY

A. Research Framework

During this research, we have used Pytorch which is an optimized tensor library, used for natural language processing and deep learning. Pytorch permits its users to use data utility and customized order of data. We have also used Pandas and scikit-learn Python libraries which are heavily used in data analysis, machine learning and data-mining.

B. Dataset

We have used a dataset known as the Employment Scam Aegean Dataset (EMSCAD) [12], which could be a set of job ads that are collected for enlisting scam detection research. It contains one set of 17,880 real-life job ads. The files contain one ads per line. Every line consists of eighteen

columns. The records of EMSCAD were annotated manually and categorized into 2 categories. Moreover, the dataset holds 17,014 legitimate and 866 deceitful job ads which were disclosed in the middle of 2012 to 2014. In table I, a sample of EMSCAD dataset has been shown. From the dataset, we have selected five attributes which are title, company profile, description, requirements and benefits, which have actual text data so that we might be able to use for training. Other attributes like locations are repetitive and other attributes have binary. We have concatenated the contents of attributes with an empty space after every text. The output after merging the attributes is shown in table II.

TABLE I
SAMPLE OF DATASET

Title	Marketing Intern
company profile	We're Food52, and we've created a groundbr...
description	Food52, a fast-growing, James Beard Award-w
requirements	Experience with content management
benefits	What you will get from us...
fraudulent	f

TABLE II
COMBINED TEXTS FROM DATASET

1	Marketing Intern <h3>We're Food52, and we've c...
2	Customer Service - Cloud Video Production...

C. Data Preprocessing

The method of transforming raw data to useful and efficient format is noted as data preprocessing. One of the main forms of preprocessing is to separate unusable data. It is necessary to make computers understand the natural language as an individual understands. NLP could be a process which will process computers to grasp the language. During this work, we have established a sequential NLP that can be employed in GRU algorithms by using word tokenization, padding data and truncating data techniques. The main points of preprocessing of the data have been marked out in fig. 1.

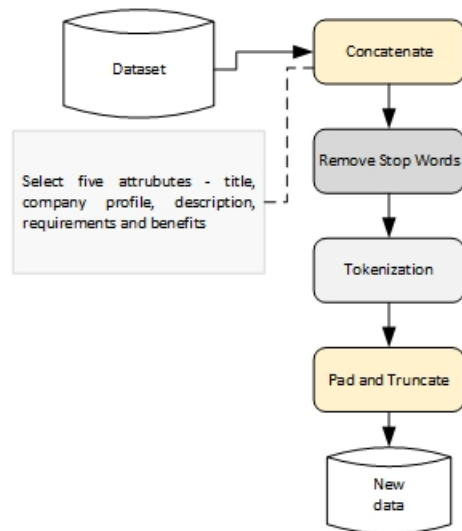


Fig. 1. Preprocessing

D. Remove Stop Words from clean texts

In Natural Language Processing (NLP), unusable words are noted as stop-words. A stop-word could be an ordinarily used word (such as “the”, “a”, “an”, “in”) [13]. While indexing entries for looking out and while recovering them as the results of a search query, a search engine ought to be programmed to ignore the stop words. We do not need stop-words to require area in our database, or consume processing time. Thus, we are able to separate them easily, by keeping a list of words that we consider stop words. NLTK (Natural Language Toolkit) in python encompasses a record of stop-words hold on in sixteen completely various languages. In our proposed model, we have used the NLTK library to store a group of stop words and filtered them out from the dataset. Before doing that we have to filter other words that have characters other than alphabets. Since the dataset we elect has both spam and ham messages, a number of the words do not have any meanings or concerned with numeric into it. During this process, we transform the data into a clean text that has solely alphabet letters.

E. Word Tokenization

Word tokenization could be a method that refers to index values that are depicted by a number. Words in a sentence are tokenized with non-repeating indexes. During the procedure, we have placed a number of vocabulary words to form word tokenizers. While forming a tokenizer, we have changed the words in a sentence into a sequence of data. The tokenizer converts the word into index and sets index to zero for unknown

words. Moreover, we make a tokenizer by setting the amount of vocabulary words to 10,000 words. We have also used this method to transform text data into a sequence of index numbers, as described in table III.

TABLE III
SAMPLE OF BEFORE AND AFTER TOKENIZATION

Before	control officer i pthe central california spca time and to function at a higher level of
After	[5934, 3069, 691, 7893, 7680, 6467, 6627, 5934, ...]

F. Padding and Truncating Data

In this section, we have made all sequences in the dataset having the same length for training using GRU algorithm. We have calculated the message length and then we have optimized it. After optimizing the length of the message, we have added zero padding in the beginning of the data sequence which has a length that is lower than the optimal length. If this current length is larger than the optimal length, we have truncated the data from the beginning until the length becomes equal to the optimal length.

G. Word Embedding

Word embedding concentrates the word vectors by the contextual data in large dataset [14]. Distributed representations are useful for the input of deep neural networks. There are some existing methods such as one-hot representation, which increases high dimensions when it comes to many categories. For our experiment, we set the word embedding dimension and the GRU hidden dimension. Then, all words embedding from the text data were embedded by 50-dimensional word vectors and the GRU hidden dimension is set to 10.

H. Gated Recurrent Unit (GRU) Model

Recurrent neural networks (RNN) and gated RNNs such as LSTM and GRU are performing well when it comes to natural language processing or speech recognition. Specially, when a model needs short and long term memories LSTMs and GRU has been preferred due to its hidden cells and gating mechanisms. The gated recurrent unit (GRU) cell has a much simpler architecture than the LSTM cell which has only two gates - reset gate and update gate. These gates are nothing

but two linear equations Reset gate is used to forget unnecessary past memories and update gate tells the model how much past memory should be taken to the next state. Fig. 2 shows the architecture and the following equations represents a GRU cell :

$$z_t = \sigma(W^{(z)}x_t + U^{(z)}h_{t-1}) \quad (1)$$

$$r_t = \sigma(W^{(r)}x_t + U^{(r)}h_{t-1}) \quad (2)$$

$$h'_t = \tanh(Wx_t + r_t \odot Uh_{t-1}) \quad (3)$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot h'_t \quad (4)$$

Here, z_t is the vectorized output of the update gate and r_t represents the reset gate. The new input x_t and the output of previous hidden cell h_{t-1} is multiplied by corresponding weights. For update gate weights and biases are $W^{(z)}$, $U^{(z)}$ and b_z and $W^{(r)}$, $U^{(r)}$ and b_r respectively. The addition of both multiplications has been passed through a sigmoid function to make the outputs lie in between 0 and 1. h'_t is a hidden state component that uses the reset gate r_t to get the relevant information from the past memory. To produce this value a hyperbolic activation function $\tanh$ has been used. Finally, the output of the current cell, h_t has been calculated using update gate z_t and hidden state component h'_t . If z_t is closer to 1, then the previous hidden value will remain unchanged and if z_t is closer to 0 then h_t will be updated by the new candidate value.

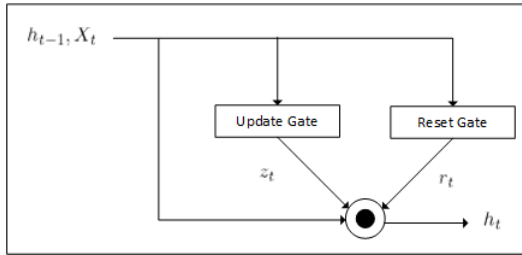


Fig. 2. Structure of GRU Cell

Our proposed model is a single layered GRU with an embedding layer and a fully connected layer. We have used sigmoid as an activation function. The embedding layer is stated before the GRU cell which we have discussed in section G. We have used two dropout layers before the embedding layer and fully connected layer to prevent the over fitting.

The network is built using Pytorch. After pre-processing the data with NLP techniques described above, we split the data into a training set 80% and validation set 20%. We created two data loaders for training and validation checks to iterate over the batches. In the training phase, we have sent a hidden state containing a 32 x 10 matrix filled with zeros in each batch. The optimizer has been set to zero to update the parameter correctly. For every mini-batch during the training phase, we typically want to explicitly set the gradients to zero before starting to do backpropagation. We calculated the loss and performed backpropagation. In backpropagation, the gradient has been clipped to avoid explosion.

I. Stochastic Gradient Descent

By combining multiple binary classifiers, stochastic gradient descent [15] are used for multi-class classification. In stochastic gradient descent, we calculate the gradient using just a random small part of the observations instead of all of them and in our model we have used 0.2 learning rate. This approach can reduce computation time.

J. Binary Cross Entropy Loss Function

Binary cross entropy compares each of the predicted probabilities to actual class output. These values can be either 0 or 1. Usually, this function calculates the score which is the differences between from the actual value.

$$H_p(q) = -\frac{1}{N} \sum_{i=1}^N y_i \cdot \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i)) \quad (5)$$

IV. RESULTS

After training the model, we have evaluated the model using the testing dataset as we discussed earlier. We have gotten the predictions as probabilities. Thus we have measured the receiver operator characteristic (ROC) curve instead of confusion matrix for better understanding about the prediction. We have gotten a 93.51% area under the ROC Curve (AUC) score with 6% training loss and 8.7% of validation loss. From the result, we can say that a simple single layered GRU has provided a better performance than some other traditional machine learning algorithms for recruitment scam classification in English language. Table IV and fig. 3 shows the training and validation losses and ROC-AUC curve of the model.

TABLE IV
AUC AND LOSS FUNCTION VALUE FOR PROPOSED MODEL

Data Type	Loss Function
Training	6%
Validation Testing	8.7%

However, the EMSCAD dataset has different features with both binary and text data but we have used only five features where each contains actual text data. Even though our proposed one-layered GRU model performed very well to classify these fraudulent recruitment emails.

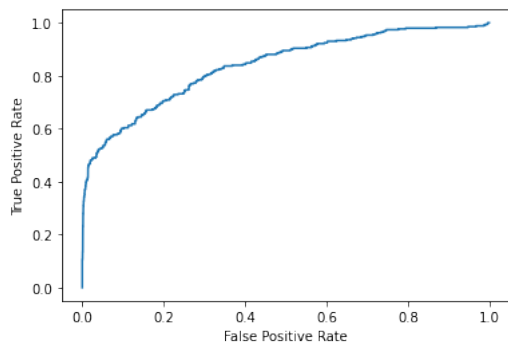


Fig. 3. The Resultant ROC AUC Curve of Our Model

V. CONCLUSION

In this research, we have proposed a GRU model to classify recruitment scams. For the experiment we have used NLP techniques for pre-processing text data into sequence using word tokenization, padding data, truncating data and word embedding technique. Finally, we have evaluated our model and the result have showed that the performance of the GRU model with 93.51% AUC score.

There is a limitation of this model which is the model cannot classify based on the information such as location, telecommuting, has company logo, has questions, employment type etc. To classify both binary and non-binary classes we would have to use different techniques rather than one model. In future we will work on a multi network analysis for the recruitment scam classification where we can use all the attributes from the dataset.

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